STA 325 Case Study Report

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Introduction

This case study explores how three factors affect the clustering of inertial particles in turbulent flows such as clouds. The data come from computer simulations: in each run, our collaborators simulated particles moving in a turbulent flow, identified clusters of particles, and measured the volume of each cluster. The dataset has 89 simulation runs, each summarizing the distribution of particle cluster volumes at one (Re, Fr, St) setting. The three parameters take the following values:

- **Reynolds number** Re: 90, 224, 398. This measures how strong the turbulence is.
- **Froude number** Fr: 0.052, 0.3, ∞. This measures the strength of gravity. Smaller Fr means stronger gravity, and $Fr = \infty$ means no gravity.
- **Stokes number** St: 14 values between 0.05 and 3. This measures particle inertia. Particles with small St follow the flow closely, while particles with large St are heavier and respond more slowly.

Our key research objectives are:

1. Predicting the mean, standard deviation, skewness, and kurtosis of the distribution at a new (Re, Fr, St) observation
2. Explaining how each parameter affects the response

Our final model is as follows, where Re_{224}, Re_{398}, $Fr_{0.3}$, and Fr_∞ are 0/1 indicators for each level (baseline $Re = 90$, $Fr = 0.052$).

$$\begin{aligned}\log \text{responses} = \beta_0 + \beta_1 \log St + \beta_2 Re_{224} + \beta_3 Re_{398} + \beta_4 Fr_{0.3} + \beta_5 Fr_\infty \\ + \beta_6 Re_{224} Fr_{0.3} + \beta_7 Re_{224} Fr_\infty + \beta_8 Re_{398} Fr_\infty + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2)\end{aligned}$$

A separate model is fit for each of the 4 response variables, so each one has its own coefficients. There is no $Re_{398} Fr_{0.3}$ term because the dataset has no observations at $Re = 398$, $Fr = 0.3$.

This model addresses both objectives. For prediction, it achieves a LOOCV MSE of 0.122 when averaged across the 4 response variables. For inference, it is a linear model, so each coefficient is easy to interpret and understand.

Methodology

Responses and predictors

Each observation in the dataset reports the first four raw moments of the response distribution, which we converted to central moments with `raw2central`. From these we computed the mean, standard deviation, skewness, and kurtosis of each distribution, which are the 4 desired response variables.

Modeling

We first compared linear modeling with generalized additive modeling (GAM). The GAM fits a separate smooth curve in $\log St$ for each (Re, Fr) combination. It had a higher LOOCV MSE than Interaction Model 2 for all 4 response variables (see the table below). We decided to use linear modeling as linear models are more easily interpreted and generally have lower variance compared to GAMs, which is valuable on unseen data.

We fitted 3 linear models for comparison. All 3 models predict the log of the response variables. The baseline model has no interaction terms. Interaction Model 1 adds interactions of $\log St$ with both Re and Fr, as well as $Re \times Fr$. Interaction Model 2 only adds $Re \times Fr$.

Averaged across the 4 response variables, the LOOCV MSE was 0.609 for the baseline model, 0.156 for Interaction Model 1, and 0.122 for Interaction Model 2. Interaction Model 1 had a slightly lower LOOCV MSE for the mean, but Interaction Model 2 performed better for the other 3 response variables, so we selected Interaction Model 2 as our final model.

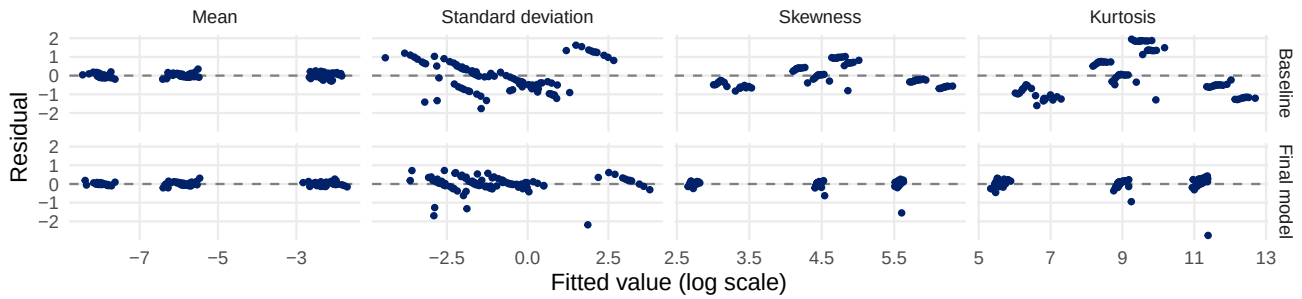
Model	Parameters	Mean	Standard deviation	Skewness	Kurtosis
Baseline (additive)	6	0.0192	0.7338	0.3561	1.3252
Interaction Model 1	13	**0.0104**	0.3290	0.0715	0.2119
Interaction Model 2	9	0.0114	**0.2565**	**0.0554**	**0.1656**
GAM	27	0.0124	0.3337	0.0639	0.1761

For the GAM, the parameter count is its effective degrees of freedom.

We log-transformed the response variables and St because they are right-skewed and span wide ranges. Re and Fr each have only 3 levels, so we treat them as categorical.

Diagnostics

The final model fits all 4 response variables well, with R^2 between 0.942 and 0.998. The plots below compare the residuals of the baseline model (top) and the final model (bottom).



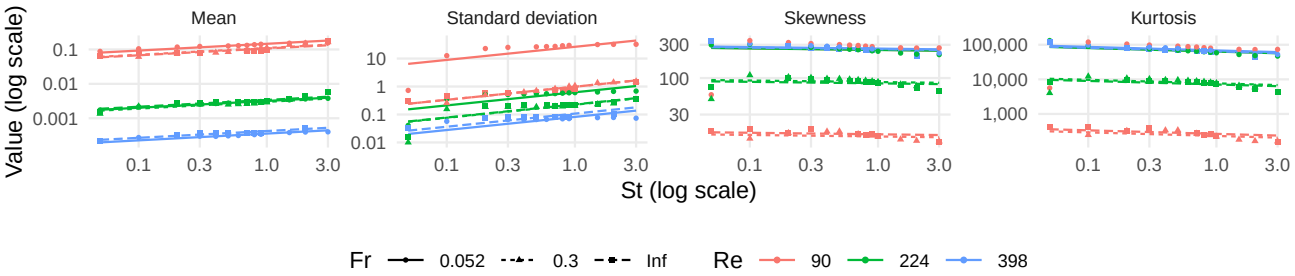


Figure 1: Fitted values from the final model (lines) and observed data (points). Both axes are on a log scale.

The baseline residuals show clear patterns, especially for skewness and kurtosis, where whole groups of observations sit above or below 0. This means the effect of Re depends on Fr. The final model removes these patterns. Its residuals are centered around 0 with roughly equal vertical spread, so the linearity and equal variance assumptions are reasonable. Some pattern remains for standard deviation.

The residuals are roughly normal, except for a few large negative residuals. These all come from observations at the smallest Stokes number, $St = 0.05$, where the model predicts values that are too high. Particles with very small St behave almost like the fluid itself and barely cluster, so the model is less accurate there. We kept these observations because there is no sign they are errors.

The dataset was collected from computer simulation runs, but observations with the same Re and Fr likely come from the same simulation, so they may not be fully independent. This could make our uncertainty estimates slightly too small.

Results

Predictions on the hold-out set

We predicted the four statistics at the 23 hold-out settings, with 95% prediction intervals computed on the log scale and then exponentiated, so point predictions estimate medians. All values are provided in `predictions.csv`. Every hold-out setting uses an (Re, Fr) combination and St value within the training range, so no prediction requires extrapolation, although the two settings with $St = 0.05$ are less reliable given the diagnostics.

The precision of the predictions varies widely across the four statistics. Predictions of the mean are precise: each interval's upper bound is only about 1.5 times its lower bound. Skewness and kurtosis intervals are wider, at about 2.5 and 4.8 times, and standard deviation intervals are the widest, at about 7.15 times, consistent with the residual pattern for standard deviation noted in the diagnostics.

Effects of each parameter

Because the model is fit on the log scale, each coefficient is a multiplicative effect: exponentiating it gives the factor by which a statistic changes. Since Re and Fr interact, the effect of gravity at a given Re is the sum of the Fr coefficient and the corresponding interaction coefficient. Figure 1 shows the fitted relationships alongside the observed data.

Turbulence intensity (Re) controls cluster size. Under strong gravity, increasing Re from 90 to 398 shrinks the mean cluster volume by a factor of about 400 and its standard deviation by a factor of about 300. Without gravity the reductions are smaller, about 250 times for the mean and 9 times for the standard deviation, but Re remains the dominant influence on the mean cluster volume. Stronger turbulence produces a wider range of eddy sizes, including much smaller ones, which appears to break particles into much smaller clusters. However, under strong gravity, Re has no detectable effect on skewness or kurtosis ($p > 0.6$).

Strong gravity creates heavy tails, but only when turbulence is weak. At $Re = 90$, moving from strong gravity ($Fr = 0.052$) to no gravity ($Fr = \infty$) reduces skewness by a factor of 16 and kurtosis by a factor of about 240. Strong gravity therefore produces a cluster-size distribution dominated by many small clusters and occasional very large ones. This effect weakens as Re increases: at $Re = 224$, removing gravity reduces kurtosis by a factor of about 9, and at $Re = 398$ it has almost no effect. One physical interpretation is that as turbulence strengthens, turbulent velocity fluctuations overwhelm gravitational settling, so the flow rather than gravity determines how particles cluster.

Moderate gravity behaves like no gravity. In all four models, the coefficients for $Fr = 0.3$ and $Fr = \infty$ are very similar. Within the range studied, gravity affects clustering only when it is strong.

Particle inertia (St) has a modest effect on scale and little effect on shape. Doubling St increases the mean cluster volume by about 15% and the standard deviation by about 38%, consistent with heavier particles being flung out of eddies and collecting into larger, more variable clusters. Doubling St lowers kurtosis by only about 7% and has no significant effect on skewness. Interaction Model 1, which let the effect of St vary by (Re, Fr), did not improve predictions, so a common St effect appears reasonable.

Conclusion

Each parameter shapes the cluster-size distribution in a different way. Turbulence intensity mostly determines how large clusters are, strong gravity mostly determines how heavy-tailed their sizes are, and particle inertia has a smaller effect on both. turbulence intensity sets how large particle clusters are, strong gravity sets how heavy-tailed their size distribution is, and particle inertia plays a secondary role. Because gravity's influence fades as turbulence strengthens, conclusions drawn at one turbulence level should not be assumed to hold at another. The model predicts the mean cluster volume reliably and the standard deviation less precisely. Its main limitations are the small design (three levels each of Re and Fr, with no runs at $Re = 398$, $Fr = 0.3$), a weaker fit at the smallest Stokes numbers, and possible correlation within simulations. New simulations at intermediate Re and Fr values and at the missing combination would address these gaps most directly.